Project Design Phase-I: Solution Architecture

Global Olympic Sports Performance Analysis Architecture Specification

Date: 19 September 2026	**Team ID:** PNT2026TMID092418
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 4 Marks

Architecture Goals & Multi-Layer System Overview

The solution architecture bridges the gap between complex historical sports data and executive decision-making across National Olympic Committees, grant agencies, and broadcast networks. It establishes a scalable 7-layer data pipeline ensuring end-to-end data integrity from raw historical records to interactive dashboards.

Layer Name	Description / Role in Architecture	Technology Used
1. Data Sources	Historical Olympic datasets containing athlete demographics, medal outcomes, and regional investment figures spanning 1992 to 2026 across 60 countries and 50 sports.	CSV Files (`athletes.csv`, `medal_results.csv`), Cloud Storage
2. Data Ingestion / Connection	Automated staging and ingestion scripts that extract raw flat files into relational cloud storage and prepare them for batch processing.	Snowflake Stages, COPY INTO Commands, Python Scripts
3. Data Preparation / Processing	Data cleansing, deduplication, handling missing values, joining athlete rosters with medal records, and mapping country codes to geographic regions.	Tableau Prep Builder, Python (pandas/numpy), Snowflake SQL
4. Data Storage / Data Layer	Centralized data warehouse layer storing structured relational tables, analytical views, and optimized data extracts for fast visual queries.	Snowflake Data Warehouse, Tableau Data Extracts (.hyper)
5. Analytics / BI Layer	Execution of core analytical logic, calculated fields, and KPI aggregations: Per-Athlete Medal Yield (Rowing: 6.916, Shooting: 6.731), Regional Efficiency % (South America: 49.64%), and Host Dominance metrics.	Tableau Calculated Fields, SQL Aggregations, Python Analytics
6. Reporting / Visualization	Interactive dashboard design featuring horizontal bar charts, investment vs. medal bubble charts, historical trend lines (1992–2026), and regional pie/donut charts.	Tableau Desktop, Tableau Public / Server
7. Business Users	Key decision-makers interacting with published dashboards: Ms. Priya (NOC Strategist), Mr. Carlos (Funding Agency Analyst), and William Robert (Broadcast Journalist).	Web Browsers, Tableau Server Portals, Executive Briefings